

Mitchell Salzman

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EDUCATION

Modeling *University of Minnesota, College of Science and Engineering*
Master of Electrical Engineering
Bachelor of Electrical Engineering GPA: 3.7 / 4.0

Minneapolis, MN
Expected December 2026
May 2026

SKILLS

Modeling: Cadence Allegro | Altium | Git | LabVIEW | LT-Spice | RF | Minitab | Python | Matlab | C/C++ | Linux | Swift | PyTorch | R
Core Competencies: Experimental Design | Six Sigma Green Belt | SolidWorks | DFM | GD&T | Technical Writing | V&V Testing

EXPERIENCE

Medtronic

R&D Electrical Systems Engineering Intern

Mounds View, MN
May 2026 – August 2026

- Built a turn-key Python test platform from scratch that **replaced a manual, 4-script process**, enabling arbitrary waveform injection, sweeps, and device configuration; **de-risking \$300M next-gen leadless pacemaker project for pilot study**
- Automated ISO 141117 EMI immunity test over PyVISA/SCPI (waveform generator injection, oscilloscope capture) with one-click runs, cutting each condition from **60 min to 1 min across 24/7 runs (1,400 datasets)**.
- Developed a real-time telemetry decoder for **32 fields, validated by 18 system verification tests**.

Medtronic

R&D Hardware Engineering Intern

Tempe, AZ
May 2025 – August 2025

- Led Voice of Customer (VOC) initiatives, aligning **5 functional groups** to integrate a next-gen hardware design for **deployment across multiple sites**.
- Directed a Universal IC package to replace obsolete Pin Grid Arrays, eliminating supply-chain risk, **resolving a year-long IC development block**, and reducing manufacturing steps from **11 to 3, projected to save \$400K annually**.
- Designed a multi-phase reliability study and Allegro PCB Layout to qualify stacked vias, enabling PCB routing efficiency, **higher battery capacity, and improved sensing accuracy across all future implantable devices**.

Abbott

R&D Electrical Engineer Co-op

Plymouth, MN
May 2024 – December 2024

- Drove adoption of **Jira** for project management, a **250% increase in weekly issue progression** (2.3 to 5.8/week), streamlining customer and team communication.
- Engineering owner of **5 key test methods** (IEC60601, EMI/EMC, clinical use, lifetime, motor stress, HALT), providing data for First-in-Human trials.
- Debugged medical device capital equipment, medical-grade computers and PCBs (BLE/NFC), resolving functional faults.

Velocity - Teamvantage

Program Management Engineering Intern

Forest Lake, MN
November 2022 – August 2023

- **Designed 5-part inspection fixtures in SolidWorks** for medical and defense plastic-injection components

ADDITIONAL

STRIVE Medical: President & Founder

September 2025 - Present

- Founded and scaled student engineering organization from **8 to 40 members across 5 technical sub teams in 9 months**; secured **\$5,000 in grants and sponsorships**, built an **11-person leadership team for 2 academic years**.
- **Won overall top prize at MIN-Corps Founders Day**, selected from 25 competing student ventures by leading cross-functional development of SynDex, a 7-DOF admittance-controlled haptic robotic surgery training controller and simulation.
- Launched a clinical collaboration with **UMN's Visible Heart Lab and Neurology department** to develop portable EEG.

UMN IEEE: Officer

December 2023 – June 2025

- Directed team of 6 to deliver industry networking events, workshop series, charity events, campus and high school outreach

Projects:

- Vibesit Insight: Led iOS app and protocol development (SwiftUI) for a lower-limb therapy platform.
- EEG Artifact Removal: Used **ML to filter EEG artifacts**, aiding autism diagnosis with cleaner data.
- BreatheBuddy: IoT smart inhaler improving pediatric asthma management and parental monitoring.
- VO₂Go: ML model detecting ventilatory threshold from smartwatch data, **delivering lab-grade(\$300) accuracy at zero cost**.

Volunteer Experience:

- Kesem Camp Counselor: Supporting children facing parental cancer.
- CSE Ambassadors: Mentoring first-year students on academic and career goals.